

GORANTLA KAMALAKAR REDDY

AI Engineer | Multi-Agent Systems, RAG & Full-Stack Delivery

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SUMMARY

AI Engineer with production experience building multi-agent systems, RAG pipelines, and full-stack applications at Teralogic, plus a self-directed portfolio of 4 independently architected agentic systems - each with explicit safety boundaries, evaluation harnesses, and documented production-completion paths. Comfortable owning a system end-to-end: architecture, backend, frontend, database, testing, and the judgment calls around what an autonomous system should and shouldn't be allowed to do unsupervised. Hands-on fluency directing three different agentic coding tools - Claude Code, OpenAI Codex, and Gemini CLI - across full application lifecycles: architecture, implementation, debugging, and deployment.

TECHNICAL SKILLS

AI / GenAI & Agentic Systems: LangGraph · LangChain · AgentScope · Multi-Agent Orchestration · Agent Supervisor/Router Patterns · RAG · Agentic RAG · Embeddings · Retrieval Grading · Hallucination/Citation Validation · MCP · Prompt Engineering · RAGAS Evaluation · AI Evaluation Harnesses · OpenAI Responses API · OpenAI GPT-4o · Claude API

Agentic Coding Tools: Claude Code · OpenAI Codex · Gemini CLI - full-stack delivery (architecture, backend, frontend, deployment, debugging) driven end-to-end through multiple agentic coding tools, not single-shot autocomplete

Vector Databases & Retrieval: pgvector · Pinecone · Qdrant · ChromaDB · FAISS · Hybrid BM25/Vector Retrieval

Backend & APIs: Python · FastAPI · Node.js · REST API Design · JWT Authentication · Typed Request/Response Contracts · OpenAPI

Frontend: Next.js · React · TypeScript · Vite · Framer Motion · Three.js

Data & Retrieval: PostgreSQL · pgvector · asyncpg · Redis · SQLGlot (AST-based SQL validation) · Pandas · Vector Search

Safety & Production Engineering: Human-in-the-Loop Approval Gates · Bounded Agent Retry/Revision · AST-Based Input Validation · Non-Root Containers · CORS Allowlisting · Structured Logging · Correlation IDs · Webhook Signature Verification (HMAC-SHA256)

DevOps, Infra & Observability: Docker · Docker Compose · Vercel · Scheduled Cron Jobs · CI-Ready Test Suites · Pytest · Ruff · Environment-Based Configuration · OpenTelemetry (planned/documented) · Token & Context Budget Controls · Health Checks

Engineering Practice: Data Structures & Algorithms · System Design · Git · Agile/Scrum · Technical Documentation

PROFESSIONAL EXPERIENCE

Associate Software Engineer - AI & Backend, Teralogic

Sep 2025 - Present |
Bengaluru

- Independently built a 0-to-1 AI POC (Smart Tutor) for an EdTech client - student/teacher workflows and agentic features - then contributed to 2 modules of the production platform built on top of it.
- Built RBAC authentication and per-role tool boundaries across the POC's Question Generator, Smart Solver, and Knowledge Base Manager modules; an iterative RAGAS evaluation loop improved retrieval precision 22% over 3 cycles.
- Owned backend SDLC for an ITSM microservices platform: Kafka consumers, 6-provider state machines, Redis caching, 14 REST endpoints, AES-256-GCM encryption - closing 20+ compliance gaps across 4 security dimensions.
- Designed a LangGraph multi-agent document chatbot (ClickScan) with supervisor routing across 6 domain-specialist agents, hybrid BM25/vector retrieval, MongoDB checkpoint persistence, and tenant-isolated JWT/ABAC controls.
- Built HubSpot and Notion integrations with OAuth 2.0 token lifecycle management and full CRUD test coverage (resolved 27 merge conflicts, 51 build errors, 7 SAST findings); later migrated a production LangGraph chatbot to AgentScope within days of release, preserving full observability.

Backend Intern, WhatBytes

Jun 2025 - Aug 2025 |
Remote

- Diagnosed API performance bottlenecks and delivered a fix via Redis caching and query restructuring, reducing median response time ~40%.
- Built SQL-based ETL pipelines improving ML training data readiness ~35%, with full documentation.

SELECTED AI / ENGINEERING PROJECTS

Workforce AI - Voice-AI Hiring & Talent Sourcing Platform

<https://github.com/kl987456/workforce-ai-custom>

FastAPI · asyncpg · PostgreSQL · REST API · React · TypeScript · Vite · Webhooks · Vercel Cron

- Built a full-stack voice-AI recruiting platform end to end - a FastAPI/asyncpg backend and a from-scratch React/TypeScript/Vite frontend - covering an AI phone-screening assistant and a 1,000+ profile talent-search & reachout engine that place real outbound calls through a third-party Voice AI API.
- Verified every inbound webhook with an HMAC-SHA256 signature check and reconciled four separate asynchronous delivery events (call status, recording, result, summary) into one consistent call-state record in PostgreSQL.
- Designed 8 rule-based autonomous agents (auto-dial new candidates, auto-triage completed calls, cross-pipeline sync, smart retry, do-not-call compliance guard) driven entirely by the voice agent's own structured output - no secondary LLM call - plus a scheduled Vercel Cron background sweep.
- Built the frontend's data-visualization layer from scratch: an SVG radial match-graph view scaled by real match score and a Three.js landing-page hero, on a custom design system rather than a component library.

ForgeGuard - Human-Governed Autonomous Software Engineering Agent

<https://github.com/kl987456/autonomous-software-engineering-agent>

Multi-Agent Orchestration · FastAPI · JWT Authentication · PostgreSQL · Redis · Next.js · Docker · Pytest

- Built a multi-agent software engineering pipeline (FastAPI, JWT-authenticated REST API, Next.js frontend) where a supervisor agent routes tasks across repository-analysis, planning, patch-proposal, test-proposal, and code-review agents, using PostgreSQL and Redis for state.
- Enforced a human-in-the-loop approval gate before any code change is applied, with a bounded revision loop; wrote a Pytest suite covering auth, approval flow, and path-traversal attacks, plus an evaluation suite checking review-package completeness, all runnable in CI via Docker.

Aperture - Autonomous Data Analyst (NL-to-SQL)

<https://github.com/kl987456/autonomous-data-analyst>

Multi-Agent Orchestration · FastAPI · JWT Authentication · PostgreSQL · SQLGlot (AST Parsing) · Pandas · Next.js · Docker

- Built a multi-agent NL-to-SQL analyst (FastAPI backend, Next.js frontend) that turns plain-English business questions into validated, read-only SQL and evidence-backed answers, with 7 specialized agents passing typed state between steps.
- Built a SQL safety validator with SQLGlot that parses the actual query structure, not keyword matching, to block multi-statement injection and mutations, backed by a row cap and a PostgreSQL role restricted to SELECT; secured the REST API with JWT auth, CORS allowlisting, and structured logging.

ResearchForge - Parallel Multi-Agent Research System

<https://github.com/kl987456/autonomous-multi-agent-research>

Multi-Agent Orchestration · FastAPI · JWT Authentication · PostgreSQL · Redis · OpenAI API · Next.js

- Built a multi-agent research system (FastAPI + JWT, Next.js frontend, PostgreSQL, Redis, OpenAI Responses API) where a supervisor runs web, document, and database researcher agents in parallel, routes output through a fact-checker, then lets a critic send a weak draft back for one bounded revision.
- Built a versioned, golden-set evaluation harness plus a deterministic mock LLM provider so the whole system runs in CI at zero live API cost.

EDUCATION

B.Tech, Computer Science (CSE) - Kalasalingam Academy of Research & Education

2021 - 2025 | GPA: 78%

Intermediate (MPC) - Sri Chaitanya Junior College

2019 - 2021 | 94.8%

ACHIEVEMENTS & CERTIFICATIONS

- IBM ICE Day - 1st Prize (GenAI + Cybersecurity) · ICLR 2026 Research Reproduction (TurboQuant) · Oracle AI Infra Certification · Cisco Ethical Hacking Certification · Prompt Engineering (Udemy)